TELIPORT
Script Your Future
Season 4

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Case Study No. 3

Team Details

Team MM

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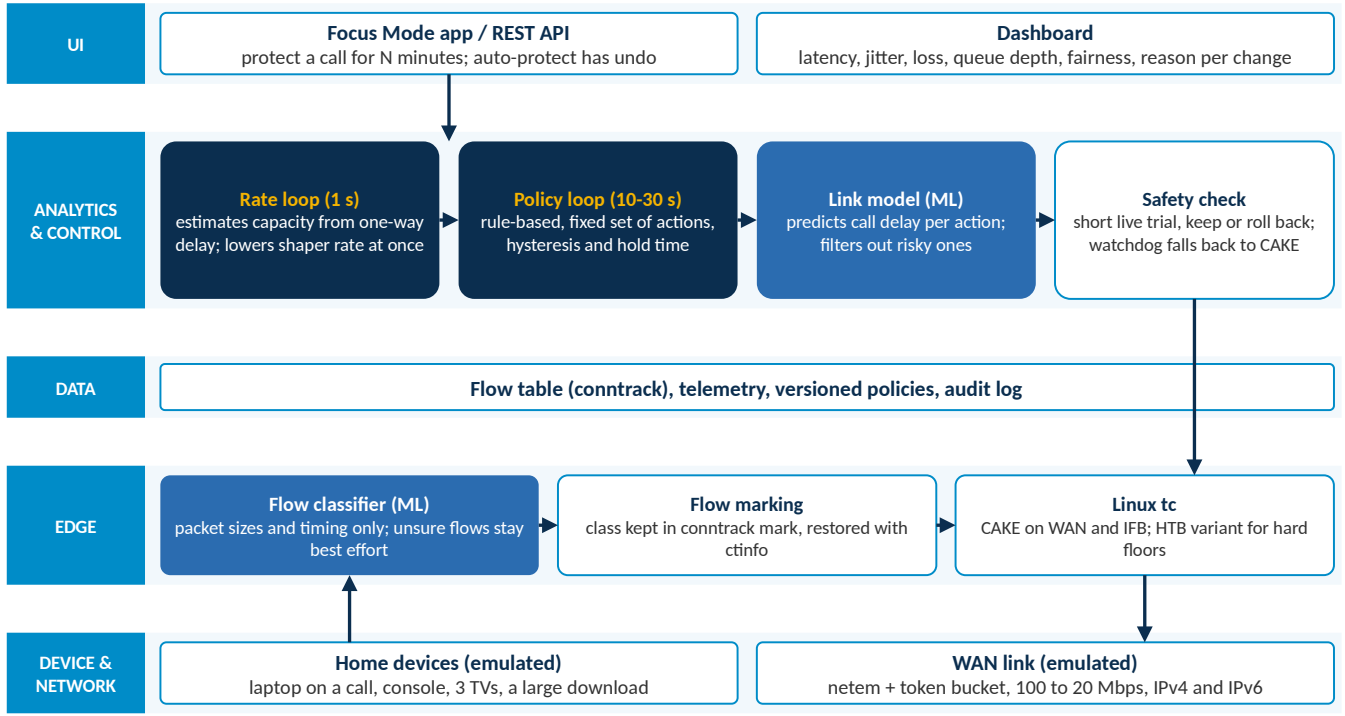
THINK. BUILD. CREATE. WIN

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PulseQoS: Adaptive Home QoS

The call never notices the download. An adaptive QoS controller built on Linux tc and CAKE, with no payload inspection.



Key design decisions

Tins: call audio and games in Voice, call video in Video (Voice is capped at 25% in diffserv4).
Trust: DSCP from the Internet is cleared, so outside traffic cannot claim priority.
Two loops: a speed drop is handled within a second; policy changes pass a safety check first.
Privacy: no payload or domain logging; the intent API is LAN-only and needs a token.

Where we use ML, and how we check it helps

1. Flow classifier: LightGBM with conformal prediction. A flow is prioritised only when the model is confident, keeping wrong boosts under a target rate (say 1%). **Compared with port rules.**

2. Link model: trained offline on lab runs; predicts how a policy change affects call delay, so risky changes are dropped before the live trial. **Measured by bad-delay seconds avoided.**

3. Auto-protect: a call starting on a busy link is protected automatically, within a daily per-device limit, with undo. **Measured by time to protection.**

Each ML part is tested against the rule-based version. If it does not improve the numbers, we drop it.

Early test we ran (27 Oct)

95th percentile ping delay during a download (ms)



Same download speed in both (18.7 of 20 Mbps): the call was protected without starving the download.
2 Linux namespaces, 20 Mbit link, 4 TCP streams, 50 pings/s, 5 runs each, no WAN delay. Round 2 compares against CAKE, CAKE autorate and cake-autorate in 14 scenarios.